

Rishabh Prajapati

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GitHub | Portfolio | LinkedIn

SUMMARY

AI/ML undergraduate at IIT Patna with hands-on experience shipping 5+ generative-AI systems, including RAG pipelines and multi-agent LLM applications, using Python, LangChain, FastAPI, and vector databases. Currently automating engineering-drawing validation for railway infrastructure as an AI Intern, cutting manual review effort by 40% through OCR- and LLM-driven data extraction. Focused on building scalable AI solutions for engineering and healthcare use cases.

EDUCATION

Degree	Institute	Year
Bachelor's Degree (AI/ML)	Indian Institute of Technology, Patna	2023–2027

EXPERIENCE

- AI Intern — Aagento** May 2026 – Present
AI Intern On-site
 - Built a **GAD extraction and validation system** using **MinerU, EasyOCR, PaddleOCR, Tesseract OCR, and MiniCPM-V**, automating extraction of engineering data from railway bridge drawings and cutting manual review time by **40%**.
 - Architected **5+ FastAPI-based rule engines** covering **Hydrology, Foundation, Span Arrangement, Level Computation, and Structural Code** validation, reducing design-check turnaround time by **50%**.
 - Implemented **3+ RAG systems** for semantic search, document retrieval, and question answering, improving retrieval accuracy by **30%** over keyword-based search.
 - Benchmarked **5 LLMs** — MiniCPM, Kimi, Ministral, Llama-3.3 Nemotron Super, and Nemotron Ultra — across 500+ engineering documents, selecting the top-performing model for production deployment.

PROJECTS

- Medical ChatBot (RAG-based Healthcare Assistant) (GitHub)**
 - Developed an AI-powered medical chatbot using RAG architecture, Gemini LLMs, and LangChain, indexing 1,000+ medical documents in Pinecone and serving answers through a Flask web interface.
 - Technologies:** RAG, Gemini LLMs, LangChain, LangSmith, Pinecone, Flask, Python, HTML/CSS, JavaScript
- AI Operations Assistant (Multi-Agent LLM System) (GitHub)**
 - Designed a multi-agent AI assistant using LangGraph and Gemini LLM that translates natural-language commands into executable actions across Planner, Executor, and Verifier agents, integrating GitHub and OpenWeather APIs with Pydantic-validated I/O.
 - Technologies:** Python, Streamlit, LangChain, LangGraph, Gemini API, GitHub API, OpenWeather API, Pydantic
- Duplicate Question Detection using NLP & ML (GitHub)**
 - Trained an NLP classification pipeline achieving 82% accuracy in identifying semantically similar questions, using text preprocessing, fuzzy matching, and a RandomForestClassifier, with results visualized via Matplotlib and Seaborn.
 - Technologies:** NLP, BeautifulSoup, FuzzyWuzzy, CountVectorizer, RandomForestClassifier, Python, Seaborn, Matplotlib
- AI DDR Report Generator (RAG-based Inspection Intelligence) (GitHub)**
 - Created an AI-driven application using RAG and Gemini LLMs to transform inspection reports into structured DDRs, extracting data from PDFs via PyMuPDF and enabling semantic search via FAISS in a Streamlit UI.
 - Technologies:** Python, Gemini, LangChain, LangGraph, FAISS, PyMuPDF, Streamlit

TECHNICAL SKILLS

- Programming:** Python, SQL
- Machine Learning & NLP:** Supervised Learning, Unsupervised Learning, Deep Learning, CNN, RNN, LSTM, Natural Language Processing
- Generative AI:** RAG, LangChain, LangGraph, Gemini API, Vector Databases (Pinecone, FAISS)
- Data Analysis:** Exploratory Data Analysis, Feature Engineering, Statistical Analysis
- Tools & Platforms:** Git, MLflow, Jupyter Notebook, Hugging Face, Docker
- Deployment:** Streamlit, Flask, FastAPI, REST APIs

POSITIONS OF RESPONSIBILITY

- Member,** Technology Club under the Student Technical Council, IIT Patna Mar. 2025 – Dec. 2025
 - Evaluated 30+ AI/ML/DL projects during the AIgnition Hackathon, judging on innovation, technical depth, and real-world applicability.

CERTIFICATIONS

- Data Science — Guvi HCL (Jun. 2025)
- Data Visualization — Kaggle (Jan. 2025)
- PyTorch Ultimate 2024: From Basics to Cutting-Edge — Udemy (Dec. 2024)
- Pandas — Kaggle (Dec. 2024)